Federal Research and Development Tax Credit Study

Gordon N. Ball, Inc.

2024

1) Executive Summary

Gordon N. Ball, Inc. ("the Company") engaged Occams Advisory ("Occams") to perform a Research and Development ("R&D") Tax Credit Study ("Study"). The Study was undertaken to analyze and document expenditures and business activities and determine qualified research expenses that may be applied towards the tax credit following the guidance of Internal Revenue Code ("IRC" or the "Code") Sections 41 and 174.

The Study covered the activities undertaken and the expenses incurred by Gordon N. Ball, Inc. during the tax year(s) 2024 (the "Study Period"). The results of the Study yielded $212,938.51 in Federal R&D tax credits.

In order to analyze, substantiate and document the Company's various R&D activities in accordance with the Code, associated regulations, and various case law, the Company collected the following information:

- General demographic information related to the Company's formation and qualification for the R&D tax credit;
- Federal and State (if applicable) tax return information;
- Employee list with job titles for the Study Period;
- Employee payroll amounts (W-2 box 1 wage equivalent) for each employee for the Study Period;
- A listing of qualified supplies, contractors, and cloud computing expenses (if applicable);
- A listing of projects completed during the Study Period;
- Time sheet data and/or estimates of qualified hours and activities; and
- In-depth project information related to qualified research activities.

Following the collection of the above information, Occams and/or its agents analyzed the data to ensure that the work completed, along with the associated expenditures qualified for the R&D tax credit by meeting the four-part-test outlined in the Code. Furthermore, Occams and/or its agents analyzed the scope of work and/or associated contractual agreements to determine if any projects would fall under the funded research exclusion of the Code or any other associated exclusions.

a) Summary of R&D Expenditures

	Tax Year 2024
Qualified Wages	$3,548,975.17
Qualified Supplies	$-
Qualified Contractors	$-
Qualified Rental or Lease Cost of Computers	$-
Total Qualified Research Expenses	$3,548,975.17
Federal R&D Tax Credit	$212,938.51

b) Summary of R&D Projects

Projects were conducted by the following employees: Bond Laupua, Eric Daniel Aguilera Acevedo, Emiliano Covarrubias, Federico Ibarra, Adriana Castillo Diaz, Jose Adan Ortiz Zaldana, Jesus Murillo Ruiz, Guadalupe Ayala Hernandez, Alejandro Aguilar, Leo Covarrubias, Lucas Hollingshead, Luis A Cruz, Josue Cruz, Martin A Ortiz, Miguel Covarrubias, Paul Torres, Omar Ramirez, Paul C. Collins, Oscar Diaz Infante, Mark Ernest Delahaye, Richard Martin Moreno Iii, Alexis Omar Lara Alvarado, Raul Hernandez Cornejo, Ricky Lipton, Robert Deondra Bell, Rogelio Alvarez, Rogelio Gomez, Amador Contreras, Christian A Martinez Ochoa, Christian Chavez, Baruch R Vargas, Cris Alejandro Chavez, Zenel Mejia, Carlos Martinez-Zepeda, Victor Antonio Gomez, Wilmer Tuz, Felipe Chable-Uc, Daniel Alejandro Guzman, Carlos Barajas Jr, David Lee Turpin, Diego Urias, Ventura Ochoa, Francisco Ceja, Giovanni Alexander Gomez, Fernando Antonio Arvizu Jr., Gabriel Natera, Adrian Guzman, Elias Salceda Maravilla, Francisco Agustin Montes, Daryl W Davies, Ernesto Rodriguez Jr, Hector Ortega Villalobos, Gustavo Martin, Gustavo Martinez Vasquez, Heisell Centeno Murrilo, Francisco Javier Gutierrez Par, Froilan Liera Cervantes Jr., David J Affonso, Jael Renteria, Frank A Hernandez, Juan Carlos Huerta Vieyra, James Dominic Machi, John J Gardner, Lido Puccetti, Julio Segura, Jose A Navarro, Javier Martinez, Jose Jesus Gil, Julian Godoy Jr, Juan Gutierrez, Neil Aaron Decker, Jose Gutierrez Tenorio, Alejandro Meza Munoz, Mitchell Jamerson, Norman Breshears, Omar J Rodriguez Ortiz, Alfonso Castaneda, Oscar Chavez Vazquez, Phone Kyaw, Miguel Tapia Jr, Porfirio Fuentes, Rafael Sanchez, Daniel Aguilera, Travis Chase Sadler, Claude Frazier, Pedro Parra, Richard Anthony Marquez, Hugo Chavez, Raul Aguilar Garcia, Tony Arthur Martindale, Jesus Moran Delatorre Mor, Jose Baldemar Gonzalez Fuentes, Jose Martinez-Mejia, Juan Maldonado, Hugo A Castaneda, Miguel Angel Garcia, Juan Dominguez, Miguel Rodriguez Galvan, Ruben Enriquez Cardoza, Porfirio A Cetina Fuentes, Victor M Becerra, Rogelio Montes Sr, Timothy Clark, Noel Hernandez Cornejo, Victor Manuel Ramos Aldrete, Raymond Ronald Guillory, and William A Velasco. A summary of the projects analyzed during the Study along with their qualifying status is contained in the table below.

Project Name	Qualification Status
Job 314 - Coyote Flood Management	Qualified
Job 317 - Tunitas Creek Beach Improvements	Qualified

2) Overview of Statutory Authority

a) Qualified Research

IRC § 41 provides a tax credit for conducting research activities undertaken to develop or improve a business component. Research constitutes "qualified research" under § 41(d)(1) if it is research:

(1) With respect to which expenditures may be treated as expenses under §174;

(2) That is undertaken for the purpose of discovering information that is technological in nature;

(3) The application of which is intended to be useful in the development of a new or improved business component of the taxpayer; and

(4) Substantially all of the activities of which constitute elements of a process of experimentation that relates to a qualified purpose.

Under §174 qualified expenses are amounts paid or incurred by the taxpayer in the taxable year in connection with his trade or business as expenses which are applied towards research or experimental activity. Expenses qualify as research or experimental costs when they are incurred in the effort to resolve uncertainty in the development or improvement of a business component. Uncertainty exists if the information available to the taxpayer does not establish the capability or method for developing or improving the business component, or the appropriate design for the business component.

To satisfy the second requirement of the statute, the research must be undertaken to discover information rooted in the principles of science. This component does not require a taxpayer seeking to obtain information that exceeds, expands or refines the common knowledge of skilled professionals in a particular field of science; nor does the component require a taxpayer to succeed in the research.

The third requirement is that the application of the research be applied to the development of a new or improved business component of the taxpayer. A business component is a product, process, formula, invention, software, or technique produced in the scope of the taxpayer's trade.

Regarding the last requirement, substantially all of a taxpayer's research activities will be considered qualified research if 80 percent or more of the research activities constitute elements of a process of experimentation that relates to a qualified purpose. A process of experimentation is a process designed to evaluate one or more alternatives to achieve a result where the capability or the method of achieving that result, or the appropriate design of that result, is uncertain as of the beginning of the taxpayer's research activities. A process of experimentation must fundamentally rely on the principles of science and involve the identification of uncertainty concerning development or improvement of a business component, the identification of one or more alternatives intended to eliminate that uncertainty, and the identification and the conduct of a process of evaluating alternatives (through, for example, modeling, simulation, or a systematic process of trial and error). This process of experimentation is undertaken for a qualified purpose if it relates to a new or improved function, performance, reliability or quality of the business component.

b) Excluded Activities

IRC § 41(d) (4) outlines activities for which the tax credit is not allowed. The excluded activities that do not meet the definition of "qualified research" are as follows:

(1) Any research conducted after the beginning of commercial production of the business component.

(2) Any research related to the adaptation of an existing business component to a particular customer's requirement or need.

(3) Any research related to the reproduction of an existing business component (in whole or in part) from a physical examination of the business component itself or from plans, blueprints, detailed specifications, or publicly available information with respect to such business component.

(4) Any –

(a) efficiency survey,

(b) activity relating to management function or technique,

(c) marketing research, testing, or development (including advertising or promotions),

(d) routine data collection, or

(e) routine or ordinary testing or inspection for quality control.

(5) Except to the extent provided in regulations, any research with respect to computer software which is developed by (or for the benefit of) the taxpayer primarily for internal use by the taxpayer, other than for use in –

(a) an activity which constitutes qualified research (determined with regard to this subparagraph), or

(b) a production process with respect to which the requirements of paragraph (1) are met.

(6) Any research conducted outside of the United States, the Commonwealth of Puerto Rico, or any possession of the United States.

(7) Any research in the social sciences, arts, or humanities.

(8) Any research to the extent funded by any grant, contract, or otherwise by another person (or governmental entity).

c) Qualified Research Expenses

IRC § 41(b)(1) explains that the term "qualified research expenses" means the sum of in-house research expenses and contract research expenses. Additionally, IRC § 41(b)(2)(A) further defines in-house research expenses as:

(1) Any wages paid or incurred to an employee for qualified services performed by such employee; and

(2) Any amount paid or incurred for supplies used in the conduct of qualified research.

Furthermore, IRC § 41(b)(3)(A) defines contract research expenses as 65 percent of any amount paid or incurred by the taxpayer to any person (other than an employee of the taxpayer) for qualified research.

i) Wages

Qualified wages are those amounts paid or incurred to an employee for qualified services performed by such employee. IRC § 41(b)(2)(B) provides that the term "qualified services" means services consisting of – 1) engaging in qualified research, or 2) engaging in the direct supervision or direct support of research activities which constitute qualified research. Furthermore, if substantially all of the services performed by an individual for the taxpayer during the taxable year consists of services meeting the requirements of clause 1 or 2, the term "qualified services" means all of the services performed by such an individual for the taxpayer during the taxable year.

Treasury Regulations § 1.41-2(d) expands upon the topic of qualified wages with the following:

> Wages paid to or incurred for an employee constitute in-house research expenses only to the extent the wages were paid or incurred for qualified services performed by the employee. If an employee has performed both qualified services and nonqualified services, only the amount of wages allocated to the performance of qualified services constitutes an in- house research expense. In the absence of another method of allocation that the taxpayer can demonstrate to be more appropriate, the amount of in-house research expense shall be determined by multiplying the total amount of wages paid or incurred for the employee during the taxable year by the ratio of the total time actually spent by the employee in the performance of qualified services for the taxpayer to the total time spent by the employee in the performance of all services for the taxpayer during the taxable year.

ii) Supplies

IRC § 41(b)(2)(B) defines "supplies" as any tangible property other than 1) land or improvements to land, and 2) property of a character subject to the allowance for depreciation. Clause 2 is not satisfied unless the property is depreciable in the hands of the taxpayer (See TG

Missouri Corp. v. Commissioner, 133 TC 278, 292 (2009)).

Supply expenses are properly claimed when used in the conduct of qualified research by the taxpayer. Treasury Regulations § 1.41-2(b) (1) explains that expenditures for supplies or for the use of personal property that are indirect research expenditures or general and administrative expenses do not qualify as in- house research expenses.

iii) Contract Research Expenses

IRC § 41(b)(3)(A) defines "contract research expenses" as 65 percent of any amount paid or incurred by the taxpayer to any person (other than an employee of the taxpayer) for qualified research. Essentially, contract research expenses are amounts expended for qualified services performed by a third party.

3) R&D Tax Credit Analysis by Project

i) Basic data

Project Name: Job 314 - Coyote Flood Management

Project Years: 2024

Man hours: 24382

Contract Type: Lump sum/fixed fee

Employees Completing Research: Aaron Decker, Neil; A Castaneda, Hugo; A Cruz, Luis; Adan Ortiz Zaldana, Jose; Aguilar, Alejandro; Aguilar Garcia, Raul; Aguilera, Daniel; Agustin Montes, Francisco; A Hernandez, Frank; Alejandro Chavez, Cris; Alexander Gomez, Giovanni; Alvarez, Rogelio; A Martinez Ochoa, Christian; A Navarro, Jose; Angel Garcia, Miguel; Anthony Marquez, Richard; Antonio Gomez, Victor; A Ortiz, Martin; Arthur Martindale, Tony; A Velasco, William; Ayala Hernandez, Guadalupe; Baldemar Gonzalez Fuentes, Jose; Barajas Jr, Carlos; Breshears, Norman; Carlos Huerta Vieyra, Juan; Castaneda, Alfonso; Castillo Diaz, Adriana; C. Collins, Paul; Ceja, Francisco; Chable-Uc, Felipe; Chase Sadler, Travis; Chavez, Christian; Chavez, Hugo; Chavez Vazquez, Oscar; Cruz, Josue; Daniel Aguilera Acevedo, Eric; Deondra Bell, Robert; Diaz Infante, Oscar; Dominguez, Juan; Enriquez Cardoza, Ruben; Fuentes, Porfirio; Godoy Jr, Julian; Gomez, Rogelio; Gutierrez, Juan; Gutierrez Tenorio, Jose; Guzman, Adrian; Hernandez Cornejo, Noel; Hernandez Cornejo, Raul; Hollingshead, Lucas; J Affonso, David; Javier Gutierrez Par, Francisco; Jesus Gil, Jose; J Rodriguez Ortiz, Omar; Laupua, Bond; Lee Turpin, David; Lipton, Ricky; Maldonado, Juan; Manuel Ramos Aldrete, Victor; Martin, Gustavo; Martinez, Javier; Martinez-Mejia, Jose; Martinez Vasquez, Gustavo; Martinez-Zepeda, Carlos; Martin Moreno Iii, Richard; M Becerra, Victor; Mejia, Zenel; Meza Munoz, Alejandro; Montes Sr, Rogelio; Moran Delatorre Mor, Jesus; Natera, Gabriel; Ochoa, Ventura; Ortega Villalobos, Hector; Parra, Pedro; Ramirez, Omar; Renteria, Jael; Rodriguez Galvan, Miguel; Rodriguez Jr, Ernesto; Ronald Guillory, Raymond; Salceda Maravilla, Elias; Sanchez, Rafael; Segura, Julio; Tapia Jr, Miguel; Torres, Paul; Tuz, Wilmer; Urias, Diego

ii) Project Description

This project aimed to design and install a sheet pile wall to control floodwaters in a creek channel within a mixed commercial and residential area. The design focused on calculating the proper embedment depth for the sheet piles and required the use of silent installation methods to minimize noise and disruption.

Specialized equipment was mounted on top of each sheet pile, using mechanical reactions to drive the piles into the ground to the specified depth. However, during installation, it was found that the calculated embedment depth was not always sufficient to provide temporary stability for the heavy equipment used, which posed challenges for safe and effective installation.

iii) Why This is a New or Improved Business Component

This project introduces novel or enhanced business components targeting performance, functionality, quality, or reliability in the following ways:

1) Innovative Solutions for Flood Wall Construction and Utility Integration

- **Process:** The activity 'Innovative Solutions for Flood Wall Construction and Utility Integration' can be classified as a new or improved process, as it embodies a systematic and organized series of actions and methods developed to address complex challenges in flood wall construction and integration with existing utilities. This process introduced area-specific analysis for sheet pile embedment, incremental stress relief cutting for flood door installation, optimized flap gate selection for low-flow drainage, on-site debris screening and soil reuse, and targeted utility reconstruction, all coordinated steps designed to achieve the project's functional goals. These advances deliver measurable improvements in performance, functionality, quality, and reliability by ensuring stable flood wall installation with reduced material use, preventing wall deformation during door installation, maintaining effective drainage for residential areas, minimizing waste through efficient debris handling, and preserving utility function with minimal disruption. Collectively, these process innovations resulted in more effective flood management and increased operational reliability, directly supporting the intended outcome of enhanced flood control and infrastructure integration.
- **Technique:** The activity 'Innovative Solutions for Flood Wall Construction and Utility Integration' can be classified as a new or improved technique because it introduced systematic, practical, and refined methods for addressing the technical challenges inherent in sheet pile wall

construction, flood door installation, flap gate functionality, debris management, and utility integration within a complex flood management project. This classification is supported by the development and application of area-specific analysis for sheet pile procurement, incremental stress relief cuts for flood doors, lighter-duty flap gate selection based on functional analysis, on-site debris screening and processing, and targeted utility reconstruction, all of which constitute systematic and organized approaches to solving technical problems. These techniques delivered measurable advances in performance, reliability, and quality by ensuring the sheet pile wall achieved required stability during silent installation, preventing wall deformation during flood door installation, eliminating nuisance drainage ponding with properly functioning flap gates, minimizing material waste through on-site debris processing, and optimizing utility penetrations to reduce unnecessary reconstruction. Collectively, these innovative techniques provided robust solutions to site-specific challenges, directly improving the functional and operational outcomes of the flood management infrastructure.

iv) Why This Project Involves Elimination of Uncertainty (Information Sought to Discover)

There is uncertainty regarding the capability, method, or design of the following technical challenges.

1) Innovative Solutions for Flood Wall Construction and Utility Integration

Capability: There was uncertainty regarding whether the specified sheet pile embedment depth would provide sufficient temporary stability for the silent installation equipment during driving operations, as the initial design did not account for the actual loads and soil conditions encountered in the field.

Capability: There was plausible uncertainty about whether the lighter-weight sheet pile sections, recalculated and procured to meet performance requirements, would still achieve the necessary flood control and structural stability under maximum anticipated flood loads.

Method: Uncertainty existed in the method of installing sheet piles using silent mechanical reactions, specifically whether the chosen technique could reliably drive piles to the required depth in variable soil conditions without causing unacceptable movement or instability.

Method: There was plausible uncertainty about whether the incremental interim cut method developed for flood door installation would effectively relieve stress and prevent wall deformation without compromising the structural integrity of the sheet pile wall.

Design: Uncertainty arose in the appropriate design of the flood doors, as making openings in stressed, interconnected sheet pile sections led to unexpected deformation due to stress release, necessitating an alternative cutting approach.

Design: There was uncertainty regarding the specified flap gate design, as the original gates required higher flow rates to operate than typically present, resulting in nuisance water accumulation and indicating that the design did not match actual drainage needs.

Capability: There was plausible uncertainty about whether the revised lighter-duty flap gates would maintain one-way flow control during all flood scenarios, given their modified operating characteristics.

Method: Uncertainty was present in the method for debris handling during excavation, as the presence and type of debris were not fully known beforehand, requiring on-site development of a screening and processing approach.

Design: Uncertainty existed in the design for utility penetrations, specifically whether the planned reconstruction of all intersecting utility lines was necessary, as the actual condition and complexity of the lines varied and some portions could be left undisturbed with a modified approach.

v) Why This Project Involves a Process of Experimentation

1) Innovative Solutions for Flood Wall Construction and Utility Integration

The activity of designing and installing the Coyote Creek flood wall required a rigorous process of experimentation to resolve multiple technical uncertainties, particularly regarding the stability and integration of the sheet pile system. The team began by identifying uncertainty in the required embedment depth for the sheet piles, as initial calculations, based on outdated Army Corps of Engineers tables, did not account for the specific soil conditions and installation forces needed for the silent driving method. To address this, detailed area-specific analyses were conducted, combining site-specific soil data with simulations and iterative calculations to model the interaction between the sheet pile sections, soil resistance, and equipment loads. This approach allowed the team to systematically test and refine embedment depths and sheet pile thicknesses, ultimately selecting lighter-weight sections that still met performance specifications while reducing material waste.

Further experimentation was evident in the construction of flood doors, where the release of accumulated stress during the cutting of sheet pile openings caused wall deformation. The team responded by developing and modeling a sequence of multiple interim cuts along the opening perimeter, using iterative trial and error to incrementally relieve stress and prevent deformation, as confirmed through on-site analysis and monitoring. For the flap gates, initial designs failed to function at low flows, so the team conducted performance analyses and simulations to evaluate alternatives, ultimately selecting and testing lighter-duty gates that performed reliably even in minimal drainage scenarios. In addressing unforeseen debris in excavation materials, a screening and processing method was devised and refined through on-site trials, ensuring clean fill could be reused efficiently. Finally, the team analyzed utility penetration challenges and modeled alternative reconstruction methods, opting for targeted reconstruction of only the exposed portions, an approach validated through careful calculation and phased implementation. Each of these solutions relied on systematic experimentation, modeling, and iterative refinement to achieve a robust, integrated flood management system.

vi) Why This Project Is Technological In Nature

1) Innovative Solutions for Flood Wall Construction and Utility Integration

Engineering:

The activity of innovative solutions for flood wall construction and utility integration was fundamentally rooted in the principle of engineering, as it required the systematic application of scientific and mathematical methods to overcome significant technical challenges associated with flood management in a complex urban environment. The team performed precise calculations and detailed, area-specific analyses that integrated existing soil data with the specified sheet pile thickness to determine optimal embedment depths, ensuring stability for the silent installation process while also optimizing material use. When the original design, which relied on outdated reference tables, led to over-specification, engineers recalculated requirements, demonstrating engineering's problem-solving approach and enabling the selection of lighter-weight sheet pile sections that maintained structural integrity. To address deformation during flood door installation, the team developed and executed a method of incremental interim cuts along the opening perimeter, an engineered solution that systematically relieved stress in the steel and prevented wall distortion. The engineering principle was further applied in the selection and installation of lighter-duty flap gates, where analysis of drainage flow requirements led to a technological adjustment that resolved the issue of nuisance water buildup in residential areas. In handling unexpected debris, the engineers innovated an on-site screening and processing method, applying practical knowledge to efficiently separate, reuse, and dispose of materials, thus minimizing waste. For utility penetrations, engineering expertise guided the proposal to reconstruct only the exposed and affected sections of utility lines, an approach that balanced functionality and safety. Each solution within the activity exemplified the essential role of engineering in addressing uncertainty, ensuring reliable performance, and achieving the required flood protection and utility integration within the project's demanding constraints.

Mathematics:

The activity 'Innovative Solutions for Flood Wall Construction and Utility Integration' was fundamentally rooted in the principle of mathematics, as the project's success depended on precise calculations, modeling, and quantitative analysis to overcome significant technical hurdles. Gordon N Ball, Inc. began by conducting detailed, site-specific mathematical analyses to determine the minimum embedment depth required for the sheet piles to ensure stability during silent installation, using available soil data and the specified sheet pile thickness to optimize procurement and avoid over-specification. This approach not only ensured the wall's structural performance but also reduced unnecessary material use. When the original design's reliance on outdated Army Corps of Engineers tables led to excessive specifications, the team recalculated and selected lighter, yet adequate, sheet pile sections, demonstrating a direct application of mathematical reasoning to engineering design and procurement. For the installation of flood doors, the team addressed the release of stress in the steel by developing a method of making multiple interim cuts inside the required opening perimeter, a solution grounded in understanding the distribution and incremental relief of stress, a mathematical concept integral to structural analysis. The redesign of flap gates relied on analyzing flow rates and drainage needs mathematically to select devices that would operate effectively even at minimal flows, preventing residential ponding. Additionally, the development of on-site screening and processing for debris, as well as the selective reconstruction of utility lines, involved calculations to minimize waste, optimize fill material use, and limit unnecessary labor. Throughout, mathematical principles were essential in modeling, evaluating, and validating each solution, enabling the team to meet or exceed functional requirements while reducing environmental impact, and ensuring the stability and reliability of the flood management infrastructure.

Physics:

The activity of developing innovative solutions for flood wall construction and utility integration was fundamentally rooted in the principles of physics, particularly in the understanding and application of forces, stress, fluid dynamics, and material behavior. The initial challenge involved calculating the correct embedment depth for sheet piles to ensure the wall could withstand specified floodwater levels, requiring a detailed analysis of soil mechanics and the interaction between the sheet pile and surrounding earth, as well as the mechanical forces generated by the silent installation equipment. Gordon N Ball, Inc. performed area-specific calculations that incorporated site soil data and the physical properties of the sheet piles, allowing for an optimized design that maintained stability during installation while reducing unnecessary material weight. During flood door construction, the team applied principles of stress and strain in steel by devising a method of making incremental cuts to gradually relieve internal stresses, thereby preventing deformation of the sheet pile wall when openings were created. For the flap gates, a critical understanding of fluid dynamics was essential; the team evaluated the flow characteristics necessary for proper gate operation and selected lighter-duty gates that responded to lower water pressures, thereby resolving drainage issues without allowing floodwater intrusion. Additionally, the solution for debris management leveraged knowledge of material separation and density, enabling on-site processing to distinguish reusable fill from waste. Lastly, the approach to utility penetrations was guided by an understanding of load distribution and structural integrity, reconstructing only the directly affected segments to maintain overall system performance while minimizing disruption. Each aspect of this activity demonstrates how leveraging physical science principles was essential to address technical uncertainties, optimize performance, and ensure the safety and reliability of the flood management system.

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i) Basic data

Project Name: Job 317 - Tunitas Creek Beach Improvements

Project Years: 2024

Man hours: 9754

Contract Type: Lump sum/fixed fee

Employees Completing Research: Aaron Decker, Neil; A Cetina Fuentes, Porfirio; Alejandro Guzman, Daniel; Alvarez, Rogelio; A Navarro, Jose; Antonio Arvizu Jr., Fernando; Baldemar Gonzalez Fuentes, Jose; Centeno Murrilo, Heisell; Chable-Uc, Felipe; Clark, Timothy; Covarrubias, Emiliano; Covarrubias, Leo; Covarrubias, Miguel; Cruz, Josue; Dominguez, Juan; Dominic Machi, James; Ernest Delahaye, Mark; Frazier, Claude; Fuentes, Porfirio; Gutierrez, Juan; Ibarra, Federico; J Affonso, David; Jamerson, Mitchell; Javier Gutierrez Par, Francisco; J Gardner, John; J Rodriguez Ortiz, Omar; Kyaw, Phone; Liera Cervantes Jr., Froilan; Martinez-Zepeda, Carlos; Mejia, Zenel; Meza Munoz, Alejandro; Murillo Ruiz, Jesus; Omar Lara Alvarado, Alexis; Puccetti, Lido; R Vargas, Baruch; Salceda Maravilla, Elias; W Davies, Daryl

ii) Project Description

This project aims to create a pedestrian path that allows people to easily reach a beach area from a nearby parking lot. The parking lot is situated next to the highway and is about 400 feet higher in elevation than the beach. The path is designed for public use, providing safe and convenient access from the parking area down to the beach.

iii) Why This is a New or Improved Business Component

This project introduces novel or enhanced business components targeting performance, functionality, quality, or reliability in the following ways:

1) Custom 3D-Fit ADA (Americans with Disabilities Act) Handrail System for Steep Terrain Pathway

- **Process:** The activity 'Custom 3D-Fit ADA Handrail System for Steep Terrain Pathway' qualifies as a new or improved process business component because it introduces a systematic and organized series of steps specifically designed to address the complex requirements of installing

ADA-compliant handrails along a steep, curving pedestrian pathway. This process delivers a significant advance in both performance and reliability by integrating GPS-based digital recording of as-built pathway alignments, transmitting precise measurements to the fabrication shop, and implementing a quality-controlled, custom fabrication and installation workflow that ensures the handrails conform exactly to the unique three-dimensional geometry of the site. By overcoming the limitations of standard prefabricated handrail sections, this approach ensures safe, accessible, and durable public access, directly enhancing the pathway's functional integrity and long-term usability for all users.

- **Technique:** The activity 'Custom 3D-Fit ADA Handrail System for Steep Terrain Pathway' qualifies as a new or improved technique, as it involved the development and application of a specific, systematic method to address the unprecedented challenge of fabricating and installing handrails along a pathway with complex three-dimensional curves and grade changes. Unlike standard methods that rely on prefabricated, rigid sections, this technique integrated precise GPS-based digital mapping of the as-built path, custom fabrication guided by exact measurements, and direct quality control throughout the process, resulting in handrails that conformed exactly to the intricate site conditions. This technique delivers a significant advance in performance, functionality, and reliability by ensuring the handrail system provides continuous, code-compliant support and safety along a pathway with highly variable geometry, which would not have been possible with traditional prefabrication methods. The result is a handrail installation that precisely matches the pathway's unique twists, turns, and slopes, thereby guaranteeing ADA accessibility and user safety across challenging terrain, and demonstrating a systematic and practical application of technology and process innovation.

iv) Why This Project Involves Elimination of Uncertainty (Information Sought to Discover)

There is uncertainty regarding the capability, method, or design of the following technical challenges.

1) Custom 3D-Fit ADA Handrail System for Steep Terrain Pathway

Capability: There was uncertainty as to whether it would be possible to design and construct a handrail system that both met ADA compliance requirements and could be installed along a pedestrian pathway with such extreme elevation change and complex, meandering alignment on steep terrain.

Method: There was uncertainty regarding the method by which accurate three-dimensional data of the as-built pathway could be captured and translated into fabrication instructions for custom handrail sections that would precisely fit the unique curves and grades of the path.

Design: There was uncertainty concerning the appropriate design for a handrail system that could provide continuous, ADA-compliant support and safety along a pathway with intersecting horizontal and vertical curves, steep drop-offs, and variable grades, all while ensuring constructability and durability in a challenging outdoor environment.

v) Why This Project Involves a Process of Experimentation

1) Custom 3D-Fit ADA Handrail System for Steep Terrain Pathway

The development of the custom 3D-fit ADA handrail system for the steep terrain pathway at Tunitas Beach Access Improvements embodied a rigorous process of experimentation to resolve substantial technical uncertainty. At the outset, the team faced the challenge of designing a handrail that could precisely follow a complex, meandering path with frequent changes in both horizontal and vertical alignment, conditions that rendered conventional prefabricated, rigid handrail sections unusable. This uncertainty necessitated an evaluative process rooted in engineering principles, as the team needed to determine whether a customized, site-specific fabrication method could achieve ADA compliance while ensuring safety and structural integrity.

To address these challenges, the team systematically employed advanced GPS technology to capture the as-built geometry of the pathway, conducting detailed digital measurements and mapping both sides of the route. These data sets were analyzed and modeled to simulate how handrail sections would need to be fabricated to match the pathway's three-dimensional twists and slopes. The fabrication process itself was iterative and experimental, with on-site visits and continuous monitoring to test and verify the fit and accuracy of each custom section before installation. Each segment was carefully marked and tracked, allowing for trial adjustments and refinements as needed. This approach, combining digital simulation, field measurement, iterative fabrication, and close analytical oversight, ensured that the final handrail system met all ADA requirements, conformed precisely to the challenging site topography, and provided a safe, reliable solution for public access.

vi) Why This Project Is Technological In Nature

1) Custom 3D-Fit ADA Handrail System for Steep Terrain Pathway

Engineering:

The development and installation of the custom 3D-fit ADA handrail system for the steep terrain pathway fundamentally relied on engineering principles to overcome significant technical challenges posed by the site's complex geometry. The steep hillside required the pathway to incorporate both horizontal curves and vertical grade changes, resulting in a highly variable alignment that could not be addressed by conventional, prefabricated handrail sections. Engineering expertise was applied through the use of advanced GPS technology to capture the precise as-built three-dimensional alignment and grade profile of both sides of the pathway, generating accurate digital data essential for the fabrication process. This data-driven approach enabled the fabrication shop to produce handrail sections with exact curvature and slope to match the intricate pathway, a task that demanded careful translation of field measurements into manufacturable components. The project team's continuous oversight of the fabrication process, including on-site verification, ensured that each custom-marked handrail section fit seamlessly during installation, thereby maintaining ADA compliance and ensuring user safety. By systematically applying engineering methods, data acquisition, digital modeling, precision fabrication, and quality control, the team resolved uncertainties inherent in adapting safety infrastructure to challenging terrain, directly improving the functionality and reliability of the public access path.

Mathematics:

The development and installation of the custom 3D-fit ADA handrail system for the steep terrain pathway at Tunitas Beach fundamentally relied on the principle of mathematics to overcome the technical challenges presented by the site's complex topography. Due to the pathway's numerous horizontal curves and vertical grade changes, precise mathematical measurements and calculations were essential to ensure that the handrail would conform exactly to the as-built path and meet strict ADA compliance standards. Advanced GPS technology was employed to digitally capture the exact three-dimensional alignment and grade profile of both sides of the pathway, generating a highly detailed mathematical model of the route. These quantitative data sets were transmitted to the fabrication shop, where they served as the basis for fabricating each handrail section to exact specifications, accounting for every twist, turn, and elevation change. The mathematical rigor embedded in the measurement, modeling, and fabrication processes was crucial to producing a handrail system that fit seamlessly along the intricate pathway, providing both safety and accessibility in a setting where standard prefabricated solutions would fail to meet the project's requirements. This precise application of mathematics ensured the reliability and functionality of the finished handrail system, directly addressing the technical uncertainty posed by the steep and variable terrain.

Physics:

The development and installation of the custom 3D-fit ADA handrail system for the steep terrain pathway fundamentally relied on principles of physics, specifically those governing force, stability, and the behavior of materials under variable loading conditions. The pathway's steep and undulating profile, with horizontal curves intersecting complex vertical grades, presented significant challenges in ensuring that the handrail would provide continuous, reliable support for users, including those with disabilities. Addressing these challenges required a precise understanding of how the handrail system would interact with both the terrain and the anticipated user loads at every point along the path. Gordon N Ball, Inc. applied advanced GPS technology to capture the as-built three-dimensional alignment and grade profile of the pathway, allowing for an exact digital representation of the physical environment. This data was critical for translating the physical complexities of the site into actionable fabrication specifications, ensuring that each steel handrail segment would be structurally sound and properly anchored, regardless of the varying slopes and curves. The custom fabrication process, monitored and verified through both digital measurements and in-person inspections, ensured that the handrail sections conformed precisely to the intended geometry, thereby maintaining the required 42" height and optimal positioning for user safety throughout the entire route. This rigorous, physics-based approach to measurement, modeling, and fabrication was essential for overcoming the technical hurdles posed by the steep terrain, ultimately resulting in a handrail system that provides both the necessary support and durability for public use on a challenging hillside path.

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4) Schedule of Qualified Research Expense Line Items

Line Item	Year	Expense Type	Gross Amount	R&D Allocation %	QRE Amount
BOND LAUPUA	2024	Wages	$3,487.01	100%	$3,487.01
ERIC DANIEL AGUILERA ACEVEDO	2024	Wages	$8,130.39	100%	$8,130.39
EMILIANO COVARRUBIAS	2024	Wages	$15,487.15	100%	$15,487.15
FEDERICO IBARRA	2024	Wages	$32,666.62	100%	$32,666.62
ADRIANA CASTILLO DIAZ	2024	Wages	$77,941.22	100%	$77,941.22
JOSE ADAN ORTIZ ZALDANA	2024	Wages	$7,740.56	100%	$7,740.56
JESUS MURILLO RUIZ	2024	Wages	$5,301.30	100%	$5,301.30
GUADALUPE AYALA HERNANDEZ	2024	Wages	$29,651.23	100%	$29,651.23
ALEJANDRO AGUILAR	2024	Wages	$24,885.84	100%	$24,885.84
LEO COVARRUBIAS	2024	Wages	$22,922.07	100%	$22,922.07
LUCAS HOLLINGSHEAD	2024	Wages	$9,193.11	100%	$9,193.11
LUIS A CRUZ	2024	Wages	$69,019.46	100%	$69,019.46
JOSUE CRUZ	2024	Wages	$82,973.79	100%	$82,973.79
MARTIN A ORTIZ	2024	Wages	$56,133.48	100%	$56,133.48
MIGUEL COVARRUBIAS	2024	Wages	$27,839.45	100%	$27,839.45
PAUL TORRES	2024	Wages	$1,685.45	100%	$1,685.45
OMAR RAMIREZ	2024	Wages	$1,685.45	100%	$1,685.45
PAUL C. COLLINS	2024	Wages	$120,303.34	100%	$120,303.34
OSCAR DIAZ INFANTE	2024	Wages	$24,585.86	100%	$24,585.86
MARK ERNEST DELAHAYE	2024	Wages	$1,760.46	100%	$1,760.46
RICHARD MARTIN MORENO III	2024	Wages	$12,531.26	100%	$12,531.26
ALEXIS OMAR LARA ALVARADO	2024	Wages	$221.48	100%	$221.48
RAUL HERNANDEZ CORNEJO	2024	Wages	$651.23	100%	$651.23
RICKY LIPTON	2024	Wages	$1,685.45	100%	$1,685.45
ROBERT DEONDRA BELL	2024	Wages	$78,100.64	100%	$78,100.64
ROGELIO ALVAREZ	2024	Wages	$5,799.40	100%	$5,799.40
ROGELIO GOMEZ	2024	Wages	$79,243.62	100%	$79,243.62
AMADOR CONTRERAS	2024	Wages	$64,078.88	100%	$64,078.88

Line Item	Year	Expense Type	Gross Amount	R&D Allocation %	QRE Amount
CHRISTIAN A MARTINEZ OCHOA	2024	Wages	$64,675.16	100%	$64,675.16
CHRISTIAN CHAVEZ	2024	Wages	$7,768.08	100%	$7,768.08
BARUCH R VARGAS	2024	Wages	$44,090.64	100%	$44,090.64
CRIS ALEJANDRO CHAVEZ	2024	Wages	$7,551.92	100%	$7,551.92
ZENEL MEJIA	2024	Wages	$78,040.77	100%	$78,040.77
CARLOS MARTINEZ-ZEPEDA	2024	Wages	$57,791.00	100%	$57,791.00
VICTOR ANTONIO GOMEZ	2024	Wages	$25,369.95	100%	$25,369.95
WILMER TUZ	2024	Wages	$8,263.01	100%	$8,263.01
FELIPE CHABLE-UC	2024	Wages	$79,683.34	100%	$79,683.34
DANIEL ALEJANDRO GUZMAN	2024	Wages	$23,498.34	100%	$23,498.34
CARLOS BARAJAS JR	2024	Wages	$27,014.86	100%	$27,014.86
DAVID LEE TURPIN	2024	Wages	$1,166.86	100%	$1,166.86
DIEGO URIAS	2024	Wages	$62,896.40	100%	$62,896.40
VENTURA OCHOA	2024	Wages	$92,684.37	100%	$92,684.37
FRANCISCO CEJA	2024	Wages	$77,749.31	100%	$77,749.31
GIOVANNI ALEXANDER GOMEZ	2024	Wages	$3,157.30	100%	$3,157.30
FERNANDO ANTONIO ARVIZU JR.	2024	Wages	$20,090.17	100%	$20,090.17
GABRIEL NATERA	2024	Wages	$9,508.07	100%	$9,508.07
ADRIAN GUZMAN	2024	Wages	$7,518.68	100%	$7,518.68
ELIAS SALCEDA MARAVILLA	2024	Wages	$54,048.95	100%	$54,048.95
FRANCISCO AGUSTIN MONTES	2024	Wages	$14,412.36	100%	$14,412.36
DARYL W DAVIES	2024	Wages	$99,444.17	100%	$99,444.17
ERNESTO RODRIGUEZ JR	2024	Wages	$10,588.66	100%	$10,588.66
HECTOR ORTEGA VILLALOBOS	2024	Wages	$101,207.68	100%	$101,207.68
GUSTAVO MARTIN	2024	Wages	$7,268.58	100%	$7,268.58
GUSTAVO MARTINEZ VASQUEZ	2024	Wages	$459.69	100%	$459.69
HEISELL CENTENO MURRILO	2024	Wages	$27,406.18	100%	$27,406.18
FRANCISCO JAVIER GUTIERREZ PAR	2024	Wages	$6,519.26	100%	$6,519.26
FROILAN LIERA CERVANTES JR.	2024	Wages	$4,290.66	100%	$4,290.66
DAVID J AFFONSO	2024	Wages	$3,650.71	100%	$3,650.71

Line Item	Year	Expense Type	Gross Amount	R&D Allocation %	QRE Amount
JAEL RENTERIA	2024	Wages	$43,023.03	100%	$43,023.03
FRANK A HERNANDEZ	2024	Wages	$612.94	100%	$612.94
JUAN CARLOS HUERTA VIEYRA	2024	Wages	$3,331.09	100%	$3,331.09
JAMES DOMINIC MACHI	2024	Wages	$9,045.49	100%	$9,045.49
JOHN J GARDNER	2024	Wages	$24,460.77	100%	$24,460.77
LIDO PUCCETTI	2024	Wages	$61,970.07	100%	$61,970.07
JULIO SEGURA	2024	Wages	$11,912.11	100%	$11,912.11
JOSE A NAVARRO	2024	Wages	$32,278.06	100%	$32,278.06
JAVIER MARTINEZ	2024	Wages	$116,989.60	100%	$116,989.60
JOSE JESUS GIL	2024	Wages	$25,376.50	100%	$25,376.50
JULIAN GODOY JR	2024	Wages	$191.54	100%	$191.54
JUAN GUTIERREZ	2024	Wages	$85,549.19	100%	$85,549.19
NEIL AARON DECKER	2024	Wages	$8,538.03	100%	$8,538.03
JOSE GUTIERREZ TENORIO	2024	Wages	$808.89	100%	$808.89
ALEJANDRO MEZA MUNOZ	2024	Wages	$76,530.20	100%	$76,530.20
MITCHELL JAMERSON	2024	Wages	$3,919.40	100%	$3,919.40
NORMAN BRESHEARS	2024	Wages	$97,148.42	100%	$97,148.42
OMAR J RODRIGUEZ ORTIZ	2024	Wages	$6,519.26	100%	$6,519.26
ALFONSO CASTANEDA	2024	Wages	$42,416.39	100%	$42,416.39
OSCAR CHAVEZ VAZQUEZ	2024	Wages	$11,657.11	100%	$11,657.11
PHONE KYAW	2024	Wages	$77,289.69	100%	$77,289.69
MIGUEL TAPIA JR	2024	Wages	$1,123.78	100%	$1,123.78
PORFIRIO FUENTES	2024	Wages	$70,605.43	100%	$70,605.43
RAFAEL SANCHEZ	2024	Wages	$14,127.70	100%	$14,127.70
DANIEL AGUILERA	2024	Wages	$10,297.68	100%	$10,297.68
TRAVIS CHASE SADLER	2024	Wages	$50,401.71	100%	$50,401.71
CLAUDE FRAZIER	2024	Wages	$736.42	100%	$736.42
PEDRO PARRA	2024	Wages	$100,170.40	100%	$100,170.40
RICHARD ANTHONY MARQUEZ	2024	Wages	$9,557.57	100%	$9,557.57
HUGO CHAVEZ	2024	Wages	$15,462.06	100%	$15,462.06

Line Item	Year	Expense Type	Gross Amount	R&D Allocation %	QRE Amount
RAUL AGUILAR GARCIA	2024	Wages	$73,208.80	100%	$73,208.80
TONY ARTHUR MARTINDALE	2024	Wages	$877.76	100%	$877.76
JESUS MORAN DELATORRE MOR	2024	Wages	$28,448.28	100%	$28,448.28
JOSE BALDEMAR GONZALEZ FUENTES	2024	Wages	$49,733.10	100%	$49,733.10
JOSE MARTINEZ-MEJIA	2024	Wages	$83,201.69	100%	$83,201.69
JUAN MALDONADO	2024	Wages	$11,775.10	100%	$11,775.10
HUGO A CASTANEDA	2024	Wages	$563.52	100%	$563.52
MIGUEL ANGEL GARCIA	2024	Wages	$18,375.76	100%	$18,375.76
JUAN DOMINGUEZ	2024	Wages	$40,293.14	100%	$40,293.14
MIGUEL RODRIGUEZ GALVAN	2024	Wages	$15,329.01	100%	$15,329.01
RUBEN ENRIQUEZ CARDOZA	2024	Wages	$12,133.73	100%	$12,133.73
PORFIRIO A CETINA FUENTES	2024	Wages	$8,542.38	100%	$8,542.38
VICTOR M BECERRA	2024	Wages	$111,030.62	100%	$111,030.62
ROGELIO MONTES SR	2024	Wages	$26,180.02	100%	$26,180.02
TIMOTHY CLARK	2024	Wages	$14,705.17	100%	$14,705.17
NOEL HERNANDEZ CORNEJO	2024	Wages	$1,004.86	100%	$1,004.86
VICTOR MANUEL RAMOS ALDRETE	2024	Wages	$115,483.13	100%	$115,483.13
RAYMOND RONALD GUILLORY	2024	Wages	$30,363.91	100%	$30,363.91
WILLIAM A VELASCO	2024	Wages	$2,154.33	100%	$2,154.33
Sum of 2024 Federal Expenses			**$3,548,975.17**		**$3,548,975.17**